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RNA lipid nanoparticles as efficient *in vivo* CRISPR-Cas9 gene editing tool for therapeutic target validation in glioblastoma cancer stem cells

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In vitro and *ex-vivo* target identification strategies often fail to predict *in vivo* efficacy, particularly for glioblastoma (GBM), a highly heterogenous tumor rich in resistant cancer stem cells (GSCs). An *in vivo* screening tool can improve prediction of therapeutic efficacy by considering the complex tumor microenvironment and the dynamic plasticity of GSCs driving therapy resistance and recurrence. This study proposes lipid nanoparticles (LNPs) as an efficient *in vivo* CRISPR-Cas9 gene editing tool for target validation in mesenchymal GSCs. LNPs co-delivering mRNA (mCas9) and single-guide RNA (sgRNA) were successfully formulated and optimized facilitating both *in vitro* and *in vivo* gene editing. *In vitro,* LNPs achieved up to 67 % reduction in green fluorescent protein (GFP) expression, used as a model target, outperforming a commercial transfection reagent. Intratumoral administration of LNPs in GSCs resulted in ~80 % GFP gene knock-out and a 2-fold reduction in GFP signal by day 14. This study showcases the applicability of CRISPR-Cas9 LNPs as a potential *in vivo* screening tool in GSCs, currently lacking effective treatment. By replacing GFP with a pool of potential targets, the proposed platform presents an exciting prospect for therapeutic target validation in orthotopic GSCs, bridging the gap between preclinical and clinical research.

1. Introduction

Gliomas are the most prevalent form of malignant tumors of the central nervous system [1]. Glioblastoma multiforme (GBM) is categorized as a grade IV astrocytoma [2,3] and accounts for ~70 % of newly diagnosed gliomas [4]. Despite the intensive therapy regime, including maximal surgical resection followed by concomitant temozolomide and radiotherapy [5], the overall patient survival remains short, with a median survival of 14 to 17 months [6]. Limited efficacy in GBM therapy is attributed to a plethora of resistance mechanisms, including the presence of physical barriers such as the blood-brain barrier [7], inter and intra-tumor heterogeneity [8], a persistent immunosuppressive microenvironment [9], and the presence of glioblastoma cancer stem cells (GSCs) [10]. Unlike the GBM bulk, GSCs are characterized by unique properties such as plasticity, self-renewal, uncontrolled cell proliferation, genomic instability, and dynamic genetic alterations [10], that play a crucial role in tumor initiation and progression [11], chemo- and radio-resistance [12] and tumor recurrence [13]. Given the critical role that GSCs play in the complexity of GBM, developing novel strategies for manipulating GSCs holds great promise in revolutionizing GBM therapeutic outcomes.

In the last decade, clustered regularly interspaced short palindromic repeats (CRISPR)-associated protein 9 (CRISPR-Cas9) has emerged as one of the most potent cutting-edge technologies in the field of gene editing [14]. CRISPR-Cas9 is composed of a single-guide RNA (sgRNA) and a Cas9 endonuclease. The sgRNA, which complements the DNA sequence (~20 bp) of the target gene, directs the Cas9 protein to the designated locus. Here, Cas9 generates double-strand breaks, leading to the formation of insertion-deletions (indels) that can cause disruption, modification, or restoration of the target DNA [15]. Since its discovery,

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significant progress has been made towards the applicability of CRISPR-Cas9 as a screening and therapeutic tool for cancer [16–18]. The majority of these studies have relied on viral vectors and *in vitro* cell cultures. Viral vectors are constrained by potential off-target effects, immunogenicity, and size of internal cargo limiting their clinical use while *in vitro* studies fail to fully replicate the complexity of *in vivo* tumors.

In the context of GSCs, the ability to perform gene editing studies *in vivo* to screen for potential therapeutic targets offers a more effective strategy compared to *in vitro* studies. However, *in vivo* feasibility of CRISPR-Cas9 gene editing is restricted by several factors, including poor stability of the CRISPR-Cas9 components in biological fluids, restricted cellular internalisation, the potential for off-target effects, and concerns regarding immunogenicity [15]. Lipid nanoparticles (LNPs) consisting of an ionizable lipid, cholesterol, PEG-lipid and a structural lipid, have gained great interest in the preclinical non-viral *in vivo* delivery of antisense oligonucleotides (ASOs) [19], small interfering RNA (siRNA) [20–22], messenger RNA (mRNA) [21] and plasmid DNA (pDNA) [20]. Two studies have reported successful pDNA-LNPs-mediated CRISPR-Cas9 gene editing of immune [23] and oncogenic [24] targets in GBM cells *in vitro*. Only one study achieved successful CRISPR RNA-LNPs-mediated gene-editing in orthotopic GBM tumors *in vivo* following intracranial administration [25]. The *in vivo* application of CRISPR RNA-LNPs in GSCs remains unexplored.

The aim of this study was to develop a versatile ionizable lipid-based nanoparticle platform capable of *in vivo* gene editing of GSCs of mesenchymal lineage. Performing gene-editing *in vivo* provides better prediction of the therapeutic potential of single or multiple gene knock-out compared to *in vitro* gene-editing followed by implantation of the edited subclones. LNPs encapsulating Cas9 mRNA (mCas9) and sgRNA targeting green fluorescent protein (GFP), as a model target, were formulated and characterized for their size, charge, encapsulation efficiency and serum stability. After confirming GFP knock-out in GSCs cells *in vitro*, *in vivo* knock-out was characterized in orthotopic murine GSCs tumors following a single intracranial administration. Interference of CRISPR edits (ICE) and flow cytometry analyses confirmed successful *in vivo* GFP gene knock-out in GSCs. The tool proposed in this study has the potential to expand the use of CRISPR-LNPs for *in vivo* target validation in GSCs/GBM tumors, ultimately contributing to the identification of novel therapeutic targets against resistant GSCs.

2. Materials and methods

2.1. Materials

For LNPs formulation, N-palmitoyl-sphingosine-1-[succinyl (methoxy polyethylene glycol) 2000] (C16-PEG2000) and 1, 2 dioleoyl-sn-glycero-3-phosphoethanolamine (DOPE) lipids were from Avanti Polar Lipids (USA). Dlin-MC3-DMA (MC3) was from Biorbyt (UK). Cholesterol, Triton X-100 (BP151), Amicon® ultra 0.5 ml filters MW CO 30 kDa and N-(2-Hydroxyethyl)piperazine-N′-(2-ethanesulfonic acid) (HEPES) were from Sigma-Aldrich (USA). Quant-iT™ RiboGreen™ RNA Reagent was from Invitrogen™. CleanCap Cas9 mRNA – 5moU (mCas9) was procured from TriLink Biotechnologies (USA). Two single guide RNAs were designed to target green fluorescent protein (GFP). The first sgRNA (sgGFP1) was designed using Chop Chop online tool (https://chopchop.cbu.uib.no/). The second sgRNA (sgGFP2) was reported by Fu et al. [26]. Both sequences (**Table S1**) were purchased from Synthego (USA). Forward and reverse primers were designed and purchased from Integrated DNA Technologies (USA) (**Table S2**). Luciferase-pcDNA3 plasmid (plasmid n. 18s964) was a gift from William Kaelin through Addgene (USA). For molecular biology work, EnGen Spy Cas9 NLS, Monarch PCR and DNA cleanup kit, Monarch RNA cleanup Kit, OneTaq® Hot Start 2× Master Mix, Gel Loading Dye Purple No SDS, RNase A, proteinase K, *Xba*I and HiScribe T7 ARCA mRNA Kit (with tailing) were from New England Biolabs (UK). UltraPure™ agarose and TRIzol™ reagent were from Invitrogen (UK). 10,000× GelRed was from Biotium (USA). MasterPure™ complete DNA & RNA purification kit was from Epicentre (USA). Formamide (deionized) was from Merck (DE).

Mouse NPE-IE glioblastoma cancer stem cells (GSCs), expressing GFP, luciferase (Luc), human epidermal growth factor receptor variant III (EGFRvIII) and lacking the expression of neurofibromatosis type 1 (Nf1) and phosphatase and tensin homolog (Pten) were engineered and kindly provided by Professor Steven Pollard from the University of Edinburgh [27]. For cell culture, glucose, accutase and DMEM/HAMS F12 media were from Sigma-Aldrich (USA). MEM Non-Essential Amino Acids (MEM NEAA), penicillin/streptomycin (P/S), bovine albumin fraction V (7.5 % solution, BSA), 2-mercaptoethanol (50 mM), phosphate-buffered saline (PBS), B-27™ supplement (50×), and N2 supplement (100×) were obtained from Gibco™ (UK). Lipofectamine™ stem transfection reagent and fetal calf serum (FCS) were purchased from Thermo Fisher Scientific (UK). Cultrex Mouse Laminin I was from Bio-Techne (USA). Recombinant Murine EGF and Recombinant Human FGF were from PeproTech (UK). For western blot analysis, RIPA buffer, cOmplete™ mini protease inhibitor cocktail and PhosSTOP™ were purchased from Sigma-Aldrich (USA). Universal antibody dilution buffer was from Merck (DE). Pierce™ BCA protein assay kit, NuPAGE™ 4 to 12 %, bis-tris, 1.0–1.5 mm, mini protein gels and SuperSignal™ west femto maximum sensitivity substrate were from Thermo Fisher Scientific (UK). Mouse anti-Cas9 (1:1000, #61978) was from Active Motif (USA). Rabbit anti-GAPDH (1:2000, #2118 L), HRP-linked anti-mouse IgG (1:2000, #7076S), HRP-linked anti-rabbit IgG (1:2000, #7074S) were from Cell Signalling (USA). For flow cytometry, Zombie Aqua™, Pacific Blue™ anti-CD45 antibody (1:200, #103126), Alexa Fluor® 647 anti-SOX2 antibody (1:200, #656108) and precision count beads were from Biolegend (USA). Formaldehyde solution (16 %) was from Thermo Fisher Scientific (UK). For animal studies, VetBond tissue adhesive was from Thermo Fisher Scientific (UK). D-luciferin Potassium Salt was from Syd Labs (USA). Iso-Vet® was from Chanelle Pharma (IE). Euthatal was from Centaur Services (UK). Mersilk non-absorbable sutures were from Aston Pharma (UK).

3. Methods

3.1. mRNA synthesis

Luciferase mRNA (mLuc) was synthesized from luciferase-pcDNA3 plasmid using *in vitro* transcription method [21]. The plasmid was first linearized with *Xba*I restriction enzyme to serve as a template for the mRNA synthesis using HiScribe T7 ARCA mRNA Kit (with tailing), following manufacturer's protocol. The concentration of RNA was quantified by NanoDrop™ One/OneC Microvolume UV–Vis Spectrophotometer (Thermo Fisher Scientific, UK).

3.2. Formulation of lipid nanoparticles (LNPs)

LNPs were formulated using ethanol injection method and following a previously established formulation to encapsulate RNAs [21]. The organic phase (**Table S3**) was prepared by dissolving the lipids: Dlin-MC3-DMA (77.86 nmol), cholesterol (103.45 nmol), DOPE (35.58 nmol), C16-PEG2000 (5.55 nmol) (35: 46.5:16:2.5 molar ratio), in 61.9 μl of ethanol. Then 6.19 μl of 20 mM citrate buffer (pH 4, in RNase free water) was added to the lipid mixture and heated at 60 °C for 10 min. The aqueous phase (**Table S3**) consisting of mCas9 and sgRNAs (5 μg total nucleic acids) was dissolved in 100 μl of 20 mM citrate buffer (pH 4, in RNase free water) and incubated at 40 °C for 10 min. Next, 10 μl of the organic phase was gradually added to the aqueous phase and mixed by pipetting for 10 s, followed by 10 s strong vortex and 20 s incubation at 40 °C. This cycle was repeated until the complete consumption of the organic phase. LNPs were incubated at 40 °C for 1 h. Residual ethanol in the sample was removed by flushing with dry nitrogen (N_2). To reduce cell toxicity when performing *in vitro* transfection, 30 kDa cut-off

Amicon® centrifugal filters were used to concentrate the samples to a quarter of the original volume. For *in vivo* studies, concentrated LNPs were resuspended in 20 mM HEPES buffer (pH ~ 7). All LNPs were formulated containing mCas9 and one of the following sgRNAs: non-targeting sgRNA (sgNeg), GFP sgRNA 1 (sgGFP1), GFP sgRNA 2 (sgGFP2), or a multi-guide mixture at 1:1 weight ratio of sgGFP1 and sgGFP2 (sgGFP1+2). mCas9 to sgRNA weight ratio was kept at 2:1 *w*/w for all conditions. Ionizable lipid to total nucleic acids mass ratio was kept 10:1 and 8:1 w/w for *in vitro* and *in vivo* studies, respectively.

3.3. Encapsulation efficiency (%EE) determination

Nucleic acids (mCas9 and sgRNAs) encapsulation efficiency was quantified using Quant-iT™ RiboGreen™ assay according to manufacturer's protocol and as previously described [28]. LNPs formulation was permeabilized by addition of 1 % Triton-X in RNase free PBS at 1:12.5 *v*/v and incubation at room temperature (RT) for 20 min under gentle shaking. For quantification of total and unentrapped nucleic acid in the final suspension, 4 μl of permeabilized LNPs were diluted with 46 μlRNase free PBS (1:12.5 *v*/v) and incubated at room temperature under gentle shaking for 20 min. This was added to an equal volume (50 μl) of Quant-iT™ RiboGreen™ RNA reagent (1:500 v/v in RNase free PBS) then measured using FLUO star OPTIMA plate reader (BMG Labtech, DE) at λex/λem 485/520 nm and 1000 gain (**Fig. S1**). Encapsulation efficiency was calculated using the following equation:

$$EE(\%) = \left(\frac{total - free}{total}\right)x100$$

3.4. Particle size, size polydispersity and zeta potential determination

Particle size (ζ-average), polydispersity index (PDI) and ζ-potential were determined by dynamic light scattering (DLS). Briefly, LNPs (10 μl) were added to 1 ml of 15 times diluted 1× PBS and transferred to a disposable folded capillary cell before measurement in a Nanosizer ZS Series (Malvern Instruments, USA). Each measurement represents the average of 20 runs performed in triplicates at 25 °C. At least 3 independent formulations were measured, results were expressed as average ± SD.

3.5. Serum stability studies

Nucleic acids solutions (mCas9 or sgRNA) or LNPs encapsulating mCas9 and a sgRNA ($LNPs_{mCas9/sgRNA}$) were either left untreated or incubated with 50 % *v*/v FCS for 0, 1, 4 or 24 h at 37 °C. Next, RNase A (0.05 μg/μl) was added and incubated for 20 min at 37 °C. RNase A was deactivated by incubation with Proteinase K (0.08 μg/μl) for 20 min at 55 °C. Preserved nucleic acids were released by addition of 10 % v/v heparin (1000 IU/ml) and incubation for 1 h at 37 °C. Released nucleic acids were purified using Monarch RNA cleanup Kit following manufacturer instructions. Purified nucleic acid samples (10 μl equivalent to ~2 μg total nucleic acid) were mixed with 6× gel loading dye (5 μl) and deionized formamide (15 μl) and incubated for 10 min at 70 °C to allow for RNA denaturation. The denatured samples were resolved on a 1.5 % denaturing agarose gel in lithium acetate borate (LAB) buffer (10 mM LAB, 0.67 % formaldehyde, pH 7) at 225 V for 25 min. Bands were visualized using ChemiDoc™ MP system (Biorad, USA).

3.6. Polymerase chain reaction (PCR)

DNA was extracted from NPE-IE cells using MasterPure Complete DNA & RNA Purification Kit following the manufacturer's instructions. PCR amplification of GFP targeted regions was performed as follows: for a 15 μl reaction, OneTaq® hot start 2× master mix (7.5 μl), primers (3 μl, 1 μM) and template DNA (1.5 μl, 10 ng/μl) were added to a PCR tube and mixed. The reaction mixture was then incubated in a T100™ Thermal Cycler (Bio-Rad) using the following settings: (1) 94 °C for 30 s, (2) 94 °C for 30 s, (3) 61 °C for 60 s, (4) 68 °C for 30 s. Steps 2 to 4 were repeated for 40 cycles, followed by 68 °C incubation for 5 min. PCR products were purified using Monarch PCR and DNA cleanup kit, following manufacturer's instructions, and quantified by NanoDrop™ One/OneC Microvolume UV–Vis Spectrophotometer.

3.7. In vitro nuclease assay

An *in vitro* nuclease assay was used to validate the targeting ability of each sgRNA alone (**Table S1**), and the combination of the two against the GFP gene. First, ribonucleic protein (RNP) complexes composed of Cas9 nuclease (1 μl, 1 μM) and sgRNAs (3 μl, 300 nM) were complexed for 10 min at room temperature in 1× Cas9 nuclease reaction buffer (23 μl). GFP DNA amplicons (3 μl, 30 nM), generated by PCR, were added to the RNP complex and incubated for 15/30/45 min at 37 °C. The final concentrations of Cas9, sgRNA and DNA were kept at 30, 30 and 3 nM, respectively. The enzymatic reaction was stopped by the addition of 6 × Gel loading dye with high EDTA content (10 mM). Digested amplicons, indicative of selective RNP activity, were resolved on a native 1.5 % *w/v* agarose gel in sodium borate (SB) buffer (10 mM sodium hydroxide and 36.5 mM boric acid, pH 8) at 225 V for 40 min (15 V/cm). Bands were visualized using ChemiDoc™ MP system.

3.8. Cell culture

NPE-IE cells were cultured in DMEM/HAM's-F12 media containing 1.5 mg/ml glucose, 1 % MEM NEAA, 1 % P/S, 1 % B27, 0.5 % N2, 0.012 % BSA and 100 μM of 50 mM 2-mercaptoethanol. Mouse laminin, murine EGF and human FGF were freshly added to the complete media prior to seeding to a final concentration of 200/10/10 ng/ml, respectively. Cells were maintained in T75 flasks in an incubator at 37 °C and 5 % CO_2. Media was changed every two to three days. Cells were detached using 1 ml of accutase for 3 min at room temperature and sub-cultured twice a week at a ratio of 1 to 10 once confluency reached 90 %.

3.9. In vitro transfection

NPE-IE cells were seeded overnight in 24-well plate at a density of 40,000 cells/well. LNPs were diluted in 500 μl of freshly prepared media to achieve a total RNA concentration of 1 or 2 μg/ml, equivalent to 15 or 30 μg of lipids per well, 0.33 or 0.66 μg of mCas9 per well, and 0.17 or 0.34 μg of sgRNA per well. Diluted LNPs were added to the cells and incubated at 37 °C, 5 % CO_2 for 48 h. Cells were then detached with accutase and sub-cultured at a 1 to 10 ratio into a 12-well plate for an additional 5 days. *In vitro* gene-editing was assessed on days 2 and/or 7 by T7 endonuclease mismatch assay and flow cytometry as described below.

3.10. T7 endonuclease mismatch assay

Two days post-transfection, DNA was extracted from harvested cells using MasterPure complete DNA and RNA purification kit and GFP target region was PCR amplified. GFP amplicons (5 μl) were directly mixed with 2 μl 10× NEB buffer 2 and 12 μl nuclease-free water. The amplicons were denatured and reannealed to form heteroduplexes (**Table S4**). T7 Endonuclease I (1 μl) was added and incubated for 15 min at 37 °C. Proteinase K (1 μl) was added and incubated for 5 min at 37 °C to inactivate T7 Endonuclease I. Digested DNA products were resolved on a 1.5 % agarose gel in 10 mM SB buffer at 225 V for 40 min. Bands were quantified using Image Lab Software. The percentage of indels formation (%) was calculated using the following formula:

$$Indels\ formation\ (\%) = \left(\frac{\sum cleaved\ product}{\sum cleaved\ product, full\ length\ amplicons}\right)x100$$

3.11. Flow cytometry of in vitro samples

Cells were harvested by accutase treatment and centrifugation at 600 xg for 3 min and then washed twice with PBS. Cells were resuspended in 200 μl and acquired using BD FACSCelesta™ flow cytometry (BD Biosciences, USA). GFP signal was acquired on the blue laser (488 nm) with 530/30 filter. Data were analyzed using FlowJo™ software (Treestar, USA). Editing efficiency is expressed as the percentage of cells that lost the expression of GFP and/or the reduction in Median Fluorescence Intensity (MFI) compared to untransfected controls. Results were analyzed using one way ANOVA.

3.12. Animals

Animal experiments carried out in this study complied with the project and personal licenses granted by the UK Home Office and in accordance with the UKCCCR Guidelines (1998). Female or male C57BL/6 mice aged between 5 and 6 weeks were either obtained by Charles River (UK) or bred at King's College London at Franklin-Wilkins Biological Services Unit. Animals were maintained at 22 °C on a 12 h light and dark cycle in Individually Ventilated Cages System (IVCS), in specific pathogen-free conditions, with *ad libitum* access to food and water.

3.13. Establishment of orthotopic GSCs tumor model

Briefly, female (18–22 g) or male (24–28 g) C57BL/6 mice were anesthetized using isoflurane. To reduce post-operative discomfort and distress, animals received a subcutaneous injection (*s.c.*) of 0.3 mg/kg of Vetergesic. The scalp was swabbed with antiseptic, and a midline incision was performed. A hole was drilled through the skull at the following coordinates: 0.5 mm anterior and 1.5 mm to the left of the bregma. NPE-IE cells (400,000 cells in 2 μl) were stereotactically (Harvard Apparatus, USA) inoculated (0.2 μl/min) using a Hamilton 5 μl syringe (Thermo Fischer Scientifics, UK) at a depth of 2.5 mm. Subsequently, the incision was sutured, and surgical adhesive was applied. Animals were kept under observation on a heating mat pad until fully recovered. Tumors successfully developed in 20 % of the injected animals. Tumor growth was monitored by bioluminescence imaging as follows: anesthetized mice received a *s.c.* injection of D-luciferin (150 mg/kg) and were imaged 12 min after injection using IVIS Lumina Series III *In Vivo* Imaging System (Perkin Elmer, USA). Tumor growth was expressed as total flux (p/s) of the bioluminescence intensity over days.

3.14. Intracranial administration of LNPs

Approximately 2–5 μl of LNPs were intracranially (*i.c.*) administered in either tumor-free brain parenchymal tissue of C57BL/6 mice, or C57BL/6 mice bearing orthotopic NPE-IE tumors following the same surgical procedure adopted during the tumor inoculation process. In the case of NPE-IE bearing C57BL/6 mice, injections were performed 10 days post-tumor inoculation. The exact nucleic acid doses and experimental endpoints are listed in the respective sections for each study.

3.15. Assessment of LNPs mediated mLuc transfection in tumor-free brain parenchymal tissue using bioluminescence imaging

LNPs encapsulating mLuc were *i.c.* administered into healthy C57BL/6 mice. The utilized mLuc doses were 3.5 μg ($n = 1$), 7.5 μg (n = 1) and 10 μg (n = 1) for the first study and 10 μg ($n = 3$) for the second study. At 4, 24, 48 and 72 h post *i.c.* administration of $LNPs_{mLuc}$, anesthetized animals were *s.c.* injected with 150 mg/kg of D-luciferin, and imaged 10 min later using IVIS Lumina Series III *In Vivo* Imaging System, to assess for the translation of mLuc into functional luciferase protein. Bioluminescence signals were plotted as total flux (p/s).

3.16. Assessment of LNPs mediated mCas9 transfection in orthotopic NPE-IE tumors using western blot analysis

LNPs encapsulating 10 μg of mCas9 ($LNPs_{mCas9}$) were *i.c.* administered in the tumorous tissues as mentioned above. Twenty-four hours post-injection, animals were humanely sacrificed, and Western blot analysis was performed to detect the expression of Cas9 protein in NPE-IE tumors.

Briefly, tumors were isolated from the rest of the brain with a scalpel, then resuspended in ice-cold RIPA buffer containing protease and phosphatase inhibitors, prepared following manufacturer's instructions, at a volume equivalent to 5 times the tissue weight. Tissues were then homogenized on ice using ultrasonic microprocessor VCX 750 (Sonics, USA) with 5 cycles of 5 s pulse 10 s pause. Homogenized tissues were centrifuged at 21,000 *g* for 30 min to pellet down the insoluble fraction. The supernatant was transferred to clean microfuge tubes. These last two steps were repeated one additional time to obtain a clear solution. Protein content was quantified using Pierce™ BCA Protein Assay Kit following manufacturer's protocol. Lysates were resolved on polyacrylamide gels and transferred on 0.2 μm nitrocellulose membrane, the nitrocellulose membrane was blocked with Universal antibody dilution buffer for 1 h at RT, washed three times with TBS containing 0.1 % Tween 20 (TBS-T 0.1 %), then incubated with primary antibodies overnight at 4 °C. Unbound primary antibody was removed by three washes with TBS-T 0.1 %. Secondary antibodies were added and incubated for 1 h under gentle rotation on a tube roller (Starlab, UK). The membrane was washed three times with TBS-T 0.1 % prior to the addition of SuperSignal™ west femto maximum Sensitivity Substrate following manufacturer's protocol. Protein bands were imaged using ChemiDoc™ MP system. Quantification of protein band intensity (int) was performed using Image Lab Software. Data are presented as relative protein expression (a.u.).

3.17. Assessment of LNPs mediated GFP knock-out in NPE-IE tumors using Inference of CRISPR Edits (ICE) and flow cytometry analysis

To assess the LNPs' ability to knock-out GFP expression following *i.c.* injection of LNPs in NPE-IE tumors, animals were euthanized 2 ($n = 4$), 7 ($n = 5$), and 14 (n = 4) days post-LNPs injection. Tumor tissues were resected using a scalpel and enzymatically digested to a single cell suspension as previously described [29]. In brief, tissues were first minced using scalpels and then digested using collagenase (1 mg/ml) and deoxyribonuclease (DNase) I (0.1 mg/ml) for 60 min at 37 °C under vigorous shaking (500 rpm). Digested tumors were then passed through a 70-μm cell strainer to collect single cells suspensions. Cells were washed twice with PBS before proceeding with DNA extraction for Inference of CRISPR Edits (ICE) analysis and antibody staining for flow cytometry as described below.

Inference of CRISPR Edits (ICE) analysis: To assess GFP gene knock-out, DNA was extracted from the single cell suspension using TRIzol™ reagent, following the manufacturer's instructions, and PCR amplicons were generated as previously mentioned. PCR amplicons (5 μl, 10–50 μg/μl) were pre-mixed with the reverse primer (5 μl, 5 μM) and sent to Genewiz/Azenta for Sanger sequencing. PCR and Sanger sequencing primers are reported in **Table S2** and **Table S5**. Sanger sequencing traces of control and edited brain tumors (.ab1 files) were analyzed using Inference of CRISPR Edits software (https://ice.synthego.com/) [30]. Data were reported as percentage of total indels, frame shift and non-frame shift indels, or gene knock-out.

Flow cytometry analysis: Single cell suspension was first stained with viability marker Zombie Aqua ™ following manufacturer's protocol. All antibodies were used at a 1:200 dilution. Surface staining was performed using Pacific Blue™ anti-mouse CD45 antibody for 20 min at room temperature, followed by two PBS washes. Cells were fixed with 4 % formaldehyde for 20 min at RT, then washed twice with PBS. Since the aim was to monitor GFP expression on NPE-IE glioblastoma stem cells, intracellular staining of stem cell transcription factor SOX2 was performed on fixed cells using True-Nuclear Transcription Factor Buffer Set and Alexa Fluor® 647 anti-SOX2 antibody, following manufacturer's instructions. Precision Count Beads (20 μl) were freshly added to the stained cells before acquisition on BD FACSCelesta™ flow cytometry. Data were analyzed using FlowJo™ software. Tumor cells were discriminated by negative gating for CD45 and positive gating for SOX2 ($CD45^-$/$SOX2^+$). Results are expressed as percentage of GFP negative (GFP^-) cells or the GFP MFI calculated within the $SOX2^+$ cell population.

3.18. Statistical analysis

Numerical data were analyzed using GraphPad Prism 9 (La Jolla, CA). Unless otherwise stated, results were analyzed using Student's *t*-test, as follows. First, the normality of the data was assessed using the Shapiro-Wilks test. Depending on the normality outcome, Student's t-test was conducted with or without Mann-Whitney post-test. Statistical significance is defined as * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ and, **** $p < 0.0001$.

4. Results

4.1. Preparation and physicochemical characterisation of LNPs encapsulating mCas9 and sgRNA

LNPs were formulated to encapsulate CRISPR RNA (mCas9/sgRNA) using the ethanol injection method, a technique previously applied for encapsulating mRNA, siRNA and pDNA [21,31]. The formation of LNPs involved the electrostatic interaction between a positively charged organic lipid phase, composed of Dlin-MC3-DMA, cholesterol, DOPE, C16-PEG2000 (at a molar ratio of 35:46.5:16:2.5), and a negatively charged aqueous phase containing CRISPR RNA. The characteristics of the resulting LNPs are presented in Table 1 and **Fig. S2**. LNPs exhibited a monodisperse system, with a PDI of ~0.140, and a particle size <150 nm. Under a pH of 7.4, LNPs carried a net surface charge of around 4 mV. The encapsulation efficiency of CRISPR RNA was approximately 88 %, as determined through the RiboGreen™ assay.

4.2. LNPs enhances mCas9 and sgRNA stability in the presence of serum

Agarose gel electrophoresis demonstrated that LNPs effectively enhanced the stability of CRISPR RNA under serum conditions (Fig. 1A, B). Unencapsulated RNA (free mCas9 or sgRNA) were not detectable after serum exposure, suggesting potential RNA degradation (**Fig. S3**). In the case of LNPs, upon exposure to 50 % serum, there was a noticeable decrease in the mCas9 signal strength as time passed. The mCas9 signal intensity gradually dropped by 67 %, 78 % and 86 % after 1, 4, and 24 h of serum exposure (Fig. 1B). In contrast, the signal intensity for sgRNA exhibited a slight change over time, experiencing only a 10 % decrease after 24 h of exposure (Fig. 1B), highlighting its greater stability compared to mCas9. DLS characterisation of LNPs exposed to 50 % FCS revealed an increase in the PDI, with approximately 4-fold higher at 1 and 4 h and 3-fold higher at 24 h (Fig. 1C). Upon incubation with serum, a marginal increase in size by ~10 nm after 24 h incubation with 50 % FCS (Fig. 1D). Upon introduction of LNPs to serum, a discernible shift was observed in the zeta potential, transitioning from a slightly positive charge (5 mV) to a negative charge (−11 mV) (Fig. 1E). This negative charge remained consistent even after 1, 4, and 24 h of incubation, with average values of −12 mV, −9 mV, and − 8 mV, respectively.

Table 1
Physicochemical characterisation and RNA encapsulation efficiency of LNPs formulation.

Size (d.nm)[a,d]	PDI[a,d]	Charge (mV)[b,d]	mCas9/sgRNA EE (%)[c,d]
140.61 ± 10.05	0.142 ± 0.03	4.5 ± 2.1	88.14 ± 3.88

4.3. sgRNAs designed against GFP are functional and specific

To evaluate the functionality and specificity of the designed sgRNAs, an *in vitro* nuclease assay was performed. When RNP complexes were incubated with PCR amplicons of the target gene, active and specific sgRNA should guide Cas9 to cleave the correct site, leading to the appearance of cleavage products detectable through agarose gel electrophoresis (Fig. 2A). Each sgRNA was tested individually (sgGFP1 or sgGFP2) as well as in combination (sgGFP1+2). A non-targeting sgRNA served as the negative control (sgNeg). Both sgGFP1 and sgGFP2 resulted in additional bands at the predicted cleavage sites 217/25 bp for sgGFP1 and 129/113 bp for sgGFP2 (Fig. 2B), alongside the unedited amplicons (242 bp), demonstrating the functional and specific nature of the sgRNAs. The RNP containing sgGFP1+2 showed comparable efficiency in cleaving GFP amplicons to that of individual sgRNAs. Amplicons treated with non-targeting RNP exhibited a size similar to control GFP amplicons, and no extra cleavage products were detected.

4.4. LNPs-mediated in vitro GFP gene editing resulted in efficient GFP protein knock-out in NPE-IE cells

In preliminary studies, LNPs encapsulating mCas9/sgRNA (5:1 and 2:1 *w*/w) efficiently ($p < 0.0001$) induced GFP protein knock-out in NPE-IE GSCs (**Fig. S4**). Particularly, LNPs 2:1 outperformed LNPs 5:1 ($p < 0.0001$) and were selected as the optimal formulation. Surprisingly, Lipofectamine™ Stem reagent (Lipo) positive control failed to induce noticeable alterations in GFP protein expression (**Fig. S4**). For subsequent experiments, LNPs 2:1 encapsulating mCas9 and single (sgGFP1, sgGFP2) or dual (sgGFP1+2) sgRNA sequences were used at 1 and 2 μg/ml to transfect NPE-IE cells (Fig. 3). Gene editing was assessed through a T7 mismatch assay conducted on DNA samples harvested 2 days post-LNPs transfection. Upon denaturation and re-annealing of the DNA amplicons, heteroduplex can form between amplicons derived from edited and wild-type cells (Fig. 3B). If bigger than 1 bp, these heteroduplexes are recognized and cleaved by T7 endonuclease. Semi-quantitative agarose gel electrophoresis of digested DNA samples highlighted successful gene editing in cells transfected with single or dual sgRNAs (Fig. 3C). The percentage of indel formation for sgGFP1, sgGFP2, or sgGFP1+2, stood at ~31 %, 57 %, 49 % and 44 %, 50 %, 52 %, for the 1 μg/ml and 2 μg/ml dose, respectively (Fig. 3C). LNPs mediated GFP gene editing led to a significant reduction in GFP protein expression within 2 days post-transfection (Fig. 3E, F). The percentages of GFP^+ cells (Fig. 3G) and GFP MFI (Fig. 3H) further decreased at 7 days post-transfection (Fig. 3D), with negligible changes were observed for the sgNeg-LNPs. The average percentages of GFP^+ cells for sgGFP1, sgGFP2 and sgGFP1 + 2 were ~28 %, 26 % and 25 % for the 1 μg/ml dose and 22 %, 19 % and 17 % for the 2 μg/ml dose, respectively. The order of GFP protein knock-out was sgGFP1+2 > sgGFP2 > sgGFP1. No significant differences were recorded between 1 μg/ml and 2 μg/ml dose, so the lower dose was used in subsequent studies.

4.5. LNPs successfully delivered mRNA to tumor and non-tumor brain tissues following intra-cranial administration

Following a preliminary dose selection study (**Fig. S5**), *i.c.* injection of LNPs encapsulating mLuc ($LNPs_{mLuc}$) was performed using 10 μg of mRNA, into tumor-free brain parenchymal tissue (Fig. 4A). The luciferase expression was observed initially at 4 h post-injection, with a peak expression at 24 h corresponding to BLI values of ~$2E^{05}$ and $8E^{05}$ p/s, respectively (Fig. 4B, **Fig. S6**). Subsequently, at 48 and 72 h, luciferase expression gradually decreased (Fig. 4B). In a subsequent study, LNPs encapsulating 10 μg of mCas9 ($LNPs_{mCas9}$) were tested in orthotopic NPE-IE tumors (Fig. 4A). Western blot analysis of orthotopic NPE-IE tumors intracranially administered with 10 μg of mCas9 ($LNPs_{mCas9}$)

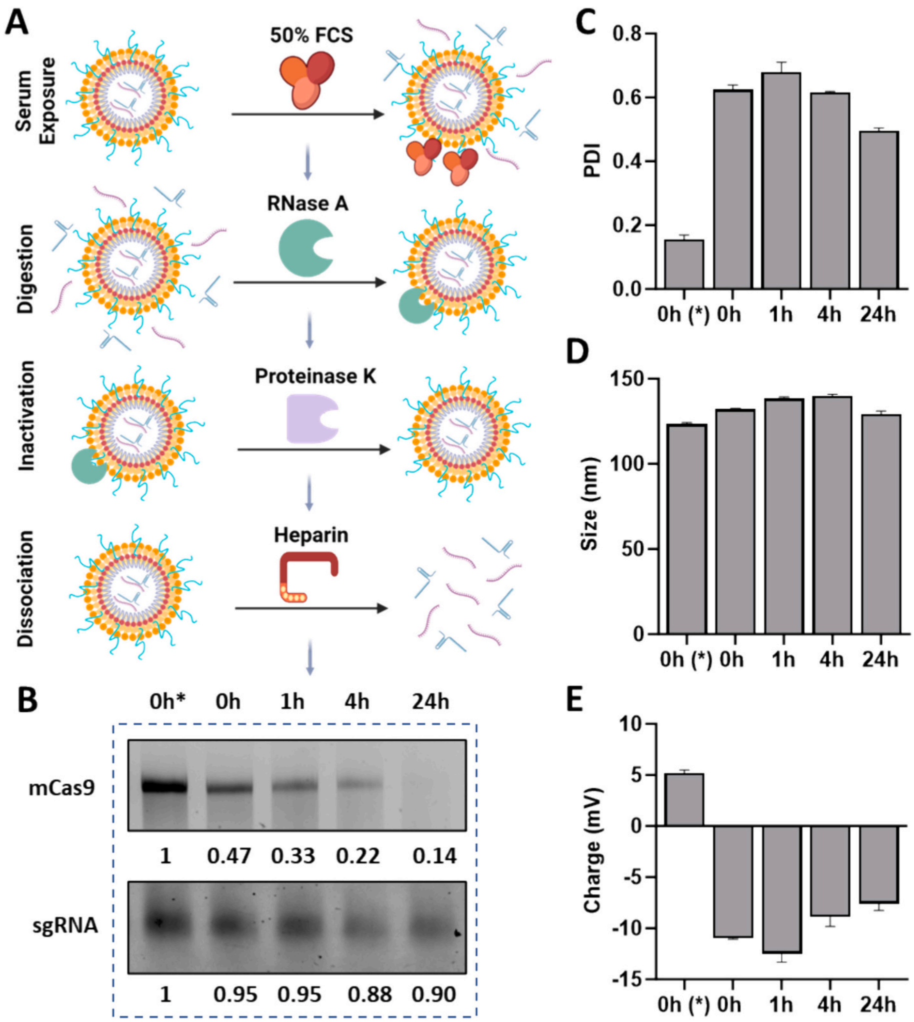

Fig. 1. LNPs enhances mCas9 and sgRNA stability in the presence of serum. Schematic representation of RNA extraction from LNPs following serum exposure **(A)**. First, LNPs were incubated with 50 % FCS for 0, 1, 4 or 24 h. LNPs were then exposed to RNase A to digest any free RNA. RNase A was inactivated by proteinase K/heating. Next, LNPs were disassembled with Heparin (10 % *v*/v) and the released nucleic acids were purified by Monarch RNA Cleanup Kit and resolved on agarose gel. Agarose gel electrophoresis of the purified RNA (~ 1 μg) loaded on 1.5 % denaturing agarose gel and imaged using ChemiDoc™ MP system **(B)**. Physiochemical characterisation of LNPs incubated in 50 % FCS for 0, 1, 4 or 24 h **(C, D, E)**. LNPs in citrate buffer not incubated with serum (*) were used as a control. Bar charts for polydispersity (PDI) **(C)**, particle size **(D)** and surface charge **(E)**. Results are expressed as means ± SD, ($n = 3$). The scheme in **A** was created with BioRender.com.

showed a significant ($p < 0.05$) increase in the relative Cas9 expression in tissue lysates collected 24 h post-LNPs administration compared to untreated controls (Fig. 4C, D, **Fig. S7**). These results confirmed that LNPs' ability to transfect mRNA in NPE-IE orthotopic tumors.

4.6. *LNPs effectively facilitates in vivo GFP gene editing in orthotopic NPE-IE tumors*

ICE analysis on Sanger sequencing DNA traces of orthotopic NPE-IE tumors intracranially administered with LNPs (10 μg mCas9/ 5 μg sgGFP1+2) revealed median indel percentages of 100 %, 33 %, and 82 % at 2-, 7-, and 14-day post-LNPs administration, respectively, underscoring LNPs' capability to mediate GFP gene editing *in vivo* (Fig. 5A, B). ICE analysis demonstrated minimal frameshift mutations on day 2, which gradually increased on days 7 and 14, with respective median values of 0 %, 55 %, and 100 % (Fig. 5C) that correlated with the median GFP knock-out values of 0 %, 23 %, and 82 %, respectively (Fig. 5D). Gene knock-out efficiency and frameshift indels were highest for 14 days followed by 7 days then 2 days.

Flow cytometry analysis of the tumors' single cell suspension suggested an increase in the $GFP^{-}/SOX2^{+}$ cell population with median percentages of $GFP^{-}/SOX2^{+}$ cells ~29 %, 40 % ($p < 0.05$), 40 % and 58 % for untreated, 2-, 7- and 14-days' time points, respectively (Fig. 5E, **Fig. S8**). Highest GFP^{-} population was achieved at 14 days followed by 7 days then 2 days mirroring ICE data. Median GFP MFI values of GFP^{-}/ $SOX2^{+}$ population decreased over time (Fig. 5F). The order of GFP MFI was untreated >7 days >2 days >14 days. These outcomes of this study underscore the efficacy of LNPs as a proficient non-viral delivery tool for *in vivo* gene editing of GSCs.

5. Discussion

GSCs are a subpopulation of cells within GBM tumors. GSCs make substantial contributions to GBM initiation, progression, and immune suppression. Their inherent plasticity results in tumor heterogeneity and adaptability to diverse therapeutic approaches, making them pivotal drivers of the therapeutic challenges encountered in GBM [32]. CRISPR-Cas9 gene editing emerges as a promising avenue for identifying and validating novel therapeutic targets within GSCs. For instance, CRISPR-Cas9 *in vitro* screenings have yielded substantial insights into pathways contributing to GSCs invasion [33], chemoresistance [34] and immune-evasion [26]. Nonetheless, performing these screenings *in vivo* could provide a more robust response, as it preserves the longitudinal evolution and plasticity of GSCs within the tumor microenvironment [35].

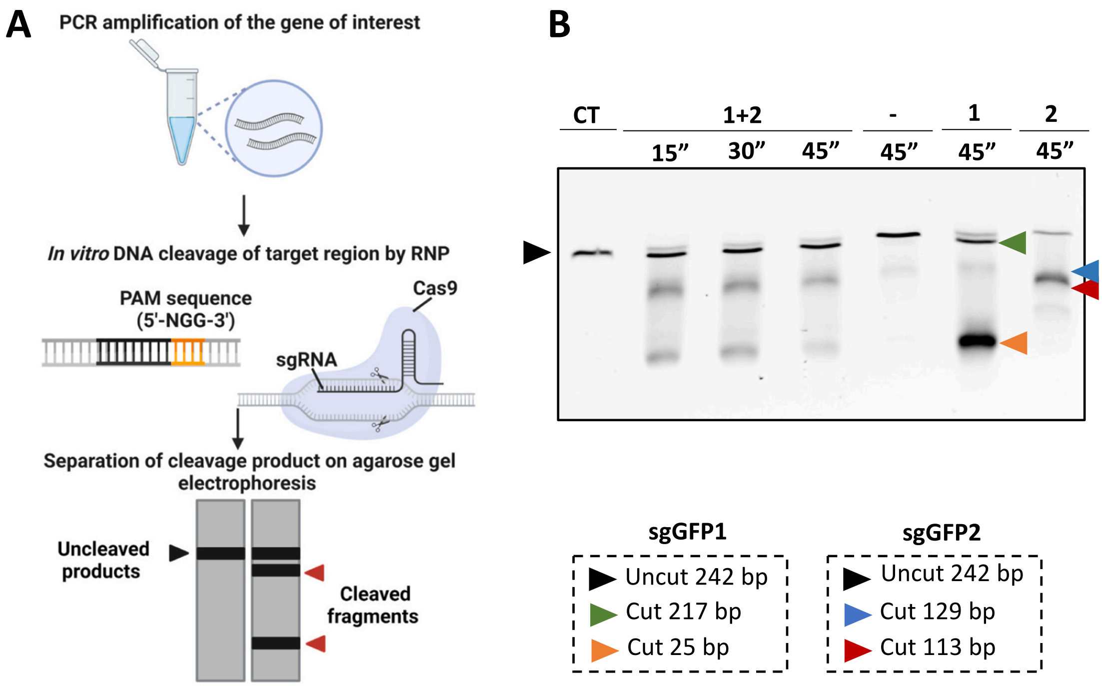

Fig. 2. SgRNAs designed against GFP are functional and specific. Schematic representation of the *in vitro* Cas9 nuclease assay used to validate sgRNAs cutting specificity **(A)**. Briefly, GFP amplicons were incubated with ribonucleic protein (RNP) complexes composed of commercial Cas9 nuclease a negative sgRNA (−) or targeting sgRNA against GFP: sgGFP1 (1), sgGFP2 (2) or a combination of the two sgGFP1+2 (1+2). Agarose gel electrophoresis of undigested control sample (CT), and digested amplicons **(B)**. Black arrows indicate full-length amplicons and colored arrows represent the cleavage products. The scheme in **A** was created with BioRender.com.

However, like many molecular approaches, the *in vivo* application of CRISPR-Cas9 faces limitations [15,36]. In this study, we employed LNPs encapsulating mCas9/sgRNA to mediate efficient *in vitro* and *in vivo* non-viral gene editing of GSCs. While in this study GFP was used as a model target, we propose the future application of CRISPR-LNPs as an *in vivo* non-viral screening tool for target identification and validation in GSCs/ GBM tumor models.

Non-viral *in vivo* gene editing for GBM has been reported using different types of nanocarriers such as LNPs [25], nanoliposomes [37], polymeric nanoparticles [38], lipid-polymer hybrids [39], and liposome-templated hydrogels [40,41] using CRISPR RNP [40,41], pDNA [37,39–41] and RNA [25]. These approaches have demonstrated efficient gene editing and highlighted their therapeutic potential for GBM treatment as a standalone intervention or in combination with other therapeutic modalities, such as temozolomide [39] or radiotherapy [37]. LNPs have demonstrated effectiveness in facilitating *in vivo* CRISPR-Cas9 delivery (including RNP, pDNA, and RNA) across different disease models such as diabetes [42,43], hypercholesterolemia [44,45], transthyretin amyloidosis [46], and cancer [25,47,48]. Only one study reported the use of LNPs for GBM gene-editing that employed localized delivery of CRISPR RNA (mCas9/sgRNA) [25]. To our knowledge, no studies have successfully performed *in vivo* CRISPR-Cas9 delivery in GSCs, as hard to transfect cells, using LNPs. The studies which have reported GSCs gene-editing *in vivo* have utilized the Cas9 protein complexed with sgRNA (RNP) encapsulated in polymeric nanocapsules [49] or exosomes [50]. The use of mRNA over plasmid and RNP is preferred to avoid immunogenicity, genomic integration concerns [36] and the presence of pre-existing immunity, respectively [51,52].

Leveraging the preferential distribution of LNPs to the liver, the progress of using CRISPR RNA-LNPs *in vivo* has primarily been explored following systemic administration, mainly targeting the liver using different ionizable lipids, including LP01, 5A2-SC8, BAMEA-O16B, 246C10 [46,53–55]. Two studies have successfully achieved gene editing in muscle tissues [56] and bone marrow [57] following intramuscular and intravenous administration of CRISPR RNA-LNPs, respectively. Assessment of CRISPR RNA-LNPs behavior in brain tumors is still limited, with a single study by Rosenblum et al. utilising L8 ionizable LNPs for intracranial administration of CRISPR RNA-LNPs in E008 GBM tumors [25]. In our research, we further optimized our previously reported MC3 ionizable-LNPs to develop CRISPR RNA-LNPs capable of successful gene editing of mesenchymal GSCs (NPE-IE) *in vitro* and *in vivo*, for the first time. Interestingly, Rosenblum et al. reported no knock-out when using MC3 lipid *in vitro,* contrary to our data. In addition to differences in other lipid compositions, differences in cell lines used could be a reason for the discrepancy in results. They did not progress MC3 to *in vivo* testing, making direct comparison with our study difficult. Our study demonstrated the superiority of LNPs over Lipofectamine™ Stem, in transfecting GSCs *in vitro*. This could be associated to differences in the uptake of LNPs compared to Lipofectamine™-RNA complexes or in the endosomal escape of the nucleic acids. Our results are in line with other reports showing that LNPs often outperform Lipofectamine™ in transfecting mRNA in various cell types, including T cells [58] and platelets [59]. LNPs could therefore be implemented as an efficient alternative *in vitro* transfection reagent of challenging cell types, such as stem cells and immune cells; serving as substitute for lipofection and electroporation methods to minimise cell stress and toxicity.

In this study we show that LNPs were able to protect mRNA for up to 4 h in the presence of 50 % serum. Considering the translation of mCas9 to Cas9 protein, 1368 amino acids (a.a.), within the cytoplasm occurs within minutes (∼5) based on a translation rate of ∼4.7 a.a. per second (a.a./s) [60], the current stability profile was considered acceptable. While our *in vitro* testing did not reveal a significant difference in GFP knock-out when using a single sgRNA *versus* a combination, we opted to use sgRNA combination for *in vivo* studies at two distinct locations

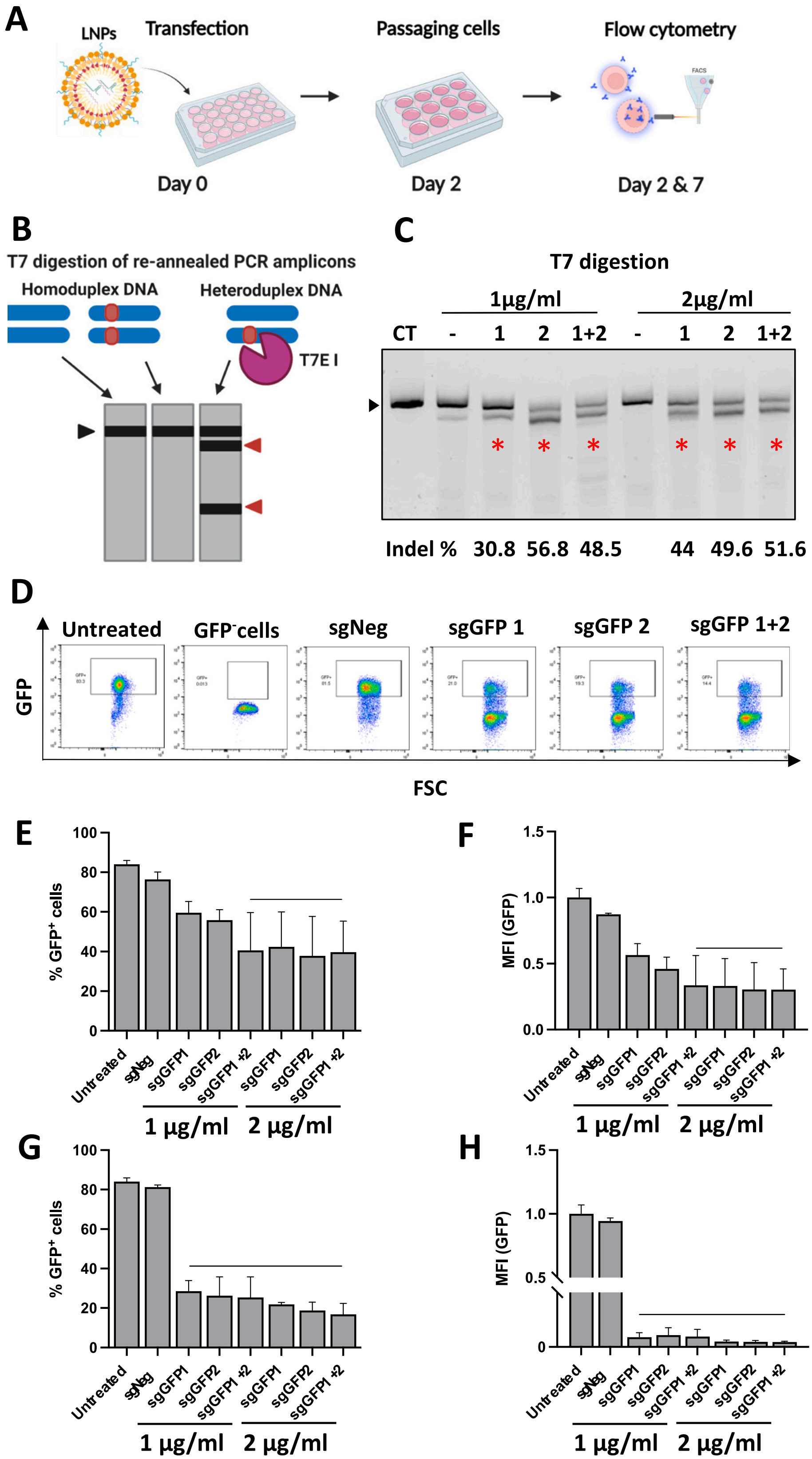

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Fig. 3. LNPs efficiently mediated *in vitro* GFP gene editing resulting in GFP protein knock-out in NPE-IE cells. Schematic representation of the experimental workflow **(A)**. NPE-IE cells were transfected with LNPs for 48 h at a concentration of 1 μg/ml or 2 μg/ml of nucleic acids composed of mCas9 and a negative sgRNA (sgNeg) or one of the targeting sgRNAs (sgGFP1, sgGFP2) or a combination of the two sgRNAs (sgGFP1+2). Gene editing was analyzed by performing T7 mismatch assay on DNA samples extracted 2- days post-transfection **(B)**. Agarose gel electrophoresis of T7 digested GFP amplicons **(C)**. The percentage (%) of indel formation was calculated using the following formula = 100 × (Σ cleaved products/Σ cleaved products, full-length GFP amplicons). Protein knock-out was measured at 2- and 7- days post-transfection using flow cytometry. Representative dot blots of untreated and GFP negative (GFP^-) control cells, and cells transfected with LNPs at 7- days post-transfection **(D)**. Percentage (%) of GFP-positive cells (GFP^+) and median fluorescence intensity (MFI) of GFP expression calculated on the entire cell population at 2- **(E, F)** and 7- **(G, H)** days post-transfection, respectively. Results are expressed as means ± SD, ($n = 3$). Significant differences were presented as * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, compared to untreated control using one-way ANOVA. Schemes in **A** and **B** were created with BioRender.com.

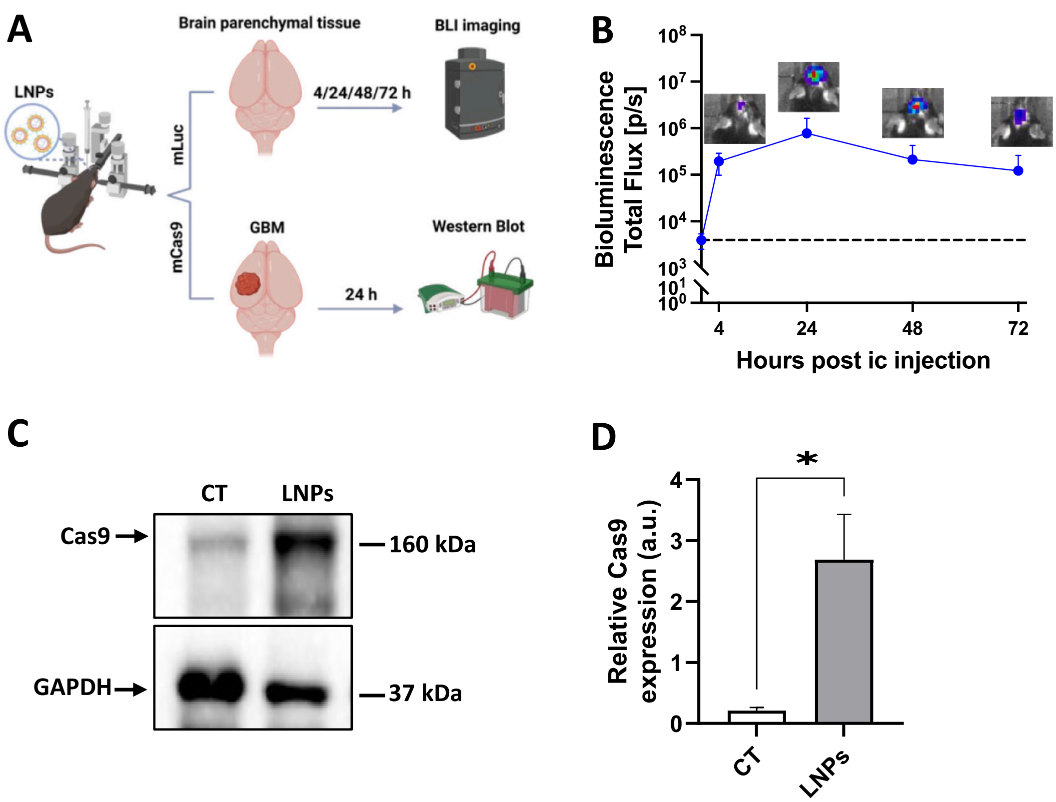

Fig. 4. Intracranial injection of LNPs effectively facilitates mLuc and mCas9 transfection in tumor-free brain parenchymal tissue and in orthotopic NPE-IE tumors. Schematic representation of the experimental workflow **(A)**. Briefly, first C57BL/6 animals were intracranially (*i.c.*) injected in tumor-free brain parenchymal tissue with LNPs encapsulating 10 μg ($n = 3$) of mRNA encoding for luciferase (mLuc). Bioluminescence Intensity (BLI) was measured to assess changes in luciferase expression at 4/24/48/72 h post-LNPs treatment. Profile of luciferase expression in animals injected with 10 μg (n = 3) of mLuc, with representative *in vivo* BLI images **(B)**. The dotted line represents the baseline signal from unrated brains. BLI is expressed as total flux (photons/s, p/s). Then, C57BL/6 mice bearing orthotopic NPE-IE tumors were *i.c.* injected with LNPs encapsulating 10 μg of mCas9 10 ($n = 7$). Tumors were extracted 24 h post-injection and protein lysate (30 μg) was immunoblotted using antibodies against Cas9 and GAPDH. Representative western blot image of untreated control (CT) and LNPs treated tumor **(C)**. Cas9 bands were quantified and normalized against GAPDH to obtain relative Cas9 expression in each tumor **(D)**. Results for bioluminescence intensity and relative Cas9 expression are expressed as means ± SD and means ± SEM, respectively. Significant differences were presented as *$p < 0.05$ using unpaired *t*-test. The scheme in **A** was created with BioRender.com.

within the gene of interest, increasing the likelihood of introducing indels and effectively disrupting the gene of interest [23]. Our data have shown promising results following intracranial administration with ~80 % GFP gene knock-out and ~2-fold reduction in GFP protein expression in GSCs ($SOX2^+$ population) by 14 days post-administration. Despite these promising results, we acknowledge variability in GFP expression among different tumors within the same group. While increasing the number of animals per group would help obtain a more accurate representation of the gene editing effect on GFP expression, we speculate that the observed variability may reflect the inherent heterogeneity and adaptability of GSCs within the tumor microenvironment [10]. Additionally, different resistance mechanisms may be responsible for this variability. For instance, Smits et al. previously reported that about a third of CRISPR-Cas9 knock-out targets retained variable levels of residual protein expression [61]. This variability was attributed to mechanisms such as translation reinitiation or exon skipping of the target gene, which could result in the retention of isoforms of the targeted protein [61].

Our study has demonstrated the versatility of the LNP platform, enabling both *in vitro* and *in vivo* CIRSPR-Cas9 screening using the same formulation. Furthermore, we have demonstrated this in a relevant GSCs model, for which there are currently no effective therapeutic targets. For future therapeutic target validation studies, intra-cranial administration offers advantages over *in vitro* gene-editing followed by cell transplantation. This approach better mimics the complex tumor microenvironment at the time of transfection including the presence of immune cells and extracellular matrix, and the intrinsic effect of LNPs on immune cell infiltration. Local administration maximises the gene editing efficacy at the site of action and spares off-target tissues. Systemic delivery requires high RNA doses and multiple administrations and may result in

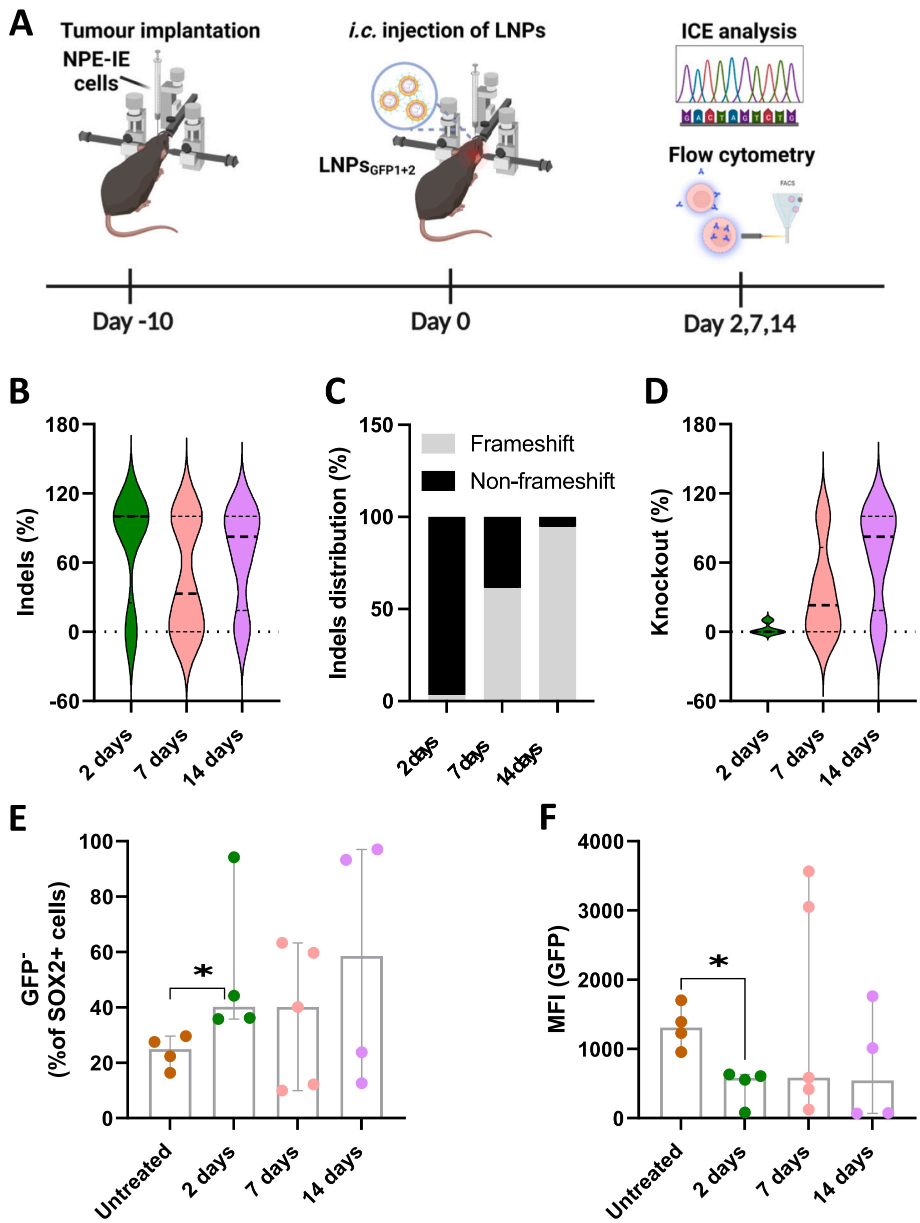

Fig. 5. Intracranial injection of LNPs effectively facilitates GFP gene editing in orthotopic NPE-IE tumors. Schematic representation of the adopted experimental procedures **(A)**. Briefly, C57BL/6 mice bearing orthotopic NPE-IE tumors were intracranially (*i.c.*) injected with LNPs encapsulating 10 μg of mCas9 and a combination of the two sgRNAs against GFP (sgGFP1+2), at a nucleic acid dose of 15 μg. On day 2 ($n = 4$), 7 ($n = 5$) and 14 ($n = 4$) post-LNPs injection tumors were processed to assess for gene editing and changes in GFP protein expression using ICE analysis **(B, C, D)** and flow cytometry (E, F), respectively. Violin plots showing the distribution of indel frequency at each time point **(B)**. Representative bar chart of frameshift and non-frameshift indel distribution expressed as percentages (%) of total indel **(C)**. Violin plots showing the distribution of gene knock-out (KO) at each time point **(D)**. Quantification of GFP negative cells (GFP^{-}) expressed as a % of the $SOX2^{+}$ population **(E)**. Median fluorescence intensity of GFP expression calculated on the $SOX2^{+}$ population **(F)**. The thick dashed line in the violin plots **B** and **D** represents the median, the thin top dashed line represents the 3rd quartile, and the bottom dashed line indicates the 1st quartile of the data set. Results in **E** and **F** are expressed as median, and error bars indicate 95 % confidence level. Each point represents an individual mouse. Significant differences were presented as $^{*}p < 0.05$ using an unpaired *t*-test, with Mann-Whitney post-test. The scheme in **A** was created with BioRender.com.

off-target side effects. By leveraging the versatility of sgRNA design, LNPs can be tailored to target any gene of interest *e.g.*, stem cell signalling pathways, onco- and immune- targets. This versatility allows for the comprehensive *in vivo* screening and validation of multiple targets. Targets identified through this *in vivo* screening approach can be further explored for clinical application through the development of new targeted therapies, small molecules, or the design of antibodies. This strategy offers a promising pathway for guiding the selection of more efficacious therapeutic approaches for GBM/GSCs.

6. Conclusions

This study highlights the potential of using LNPs to enable efficient *in vivo* CRISPR-Cas9 mediated gene in orthotopic GSCs tumors. Employing CRISPR RNA-LNP as an *in vivo* screening tool holds promise for discovering novel therapeutic targets specific to GSCs. Conducting these studies *in vivo* allow for more accurate prediction of therapeutic targets compared to *in vitro* methods, which often overlook the dynamic adaptability of GSCs during tumor progression. Future investigations will focus on the *in vivo* screening of a number of immune checkpoint targets alone or in combination with chemo- and radiotherapy. The ultimate goal is to pinpoint the therapeutic approach that holds the highest potential in effectively addressing the complexities of GSCs tumors.

CRediT authorship contribution statement

Nadia Rouatbi: Writing – original draft, Methodology, Investigation. **Adam A. Walters:** Writing – review & editing, Methodology. **Pedro M. Costa:** Methodology. **Yue Qin:** Methodology. **Revadee Liam-Or:** Methodology. **Vivien Grant:** Methodology. **Steven M. Pollard:** Writing – review & editing, Resources. **Julie Tzu-Wen Wang:** Supervision, Methodology. **Khuloud T. Al-Jamal:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jconrel.2024.09.019.

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